

Ayomide Abilewa

Electronic and Electrical Engineering student, focused on embedded systems and instrumentation

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PROFILE

Electronic and electrical engineering student at Obafemi Awolowo University, expected 2027, working across embedded systems, instrumentation and computer vision. Currently interning on Chevron Nigeria's electrical and instrumentation team. Has built sensing and detection systems from microcontroller nodes up to full-stack applications, and leads an embedded systems curriculum for secondary school students.

EDUCATION

B.Sc. Electronic and Electrical Engineering · Obafemi Awolowo University 2021–2027 (expected)
Ile-Ife, Osun State, Nigeria

EXPERIENCE

Facilities Engineering Management Intern (SIWES) · Chevron Nigeria Limited Mar 2025–Present
Lagos, Nigeria

- Rotate across the Electrical and Instrumentation (E&I) team, working on oil and gas field instrumentation and electrical systems.
- Interpret P&IDs and study control loop configurations, pneumatic signal standards (3–15 PSI), I/P converters and hazardous area classification.
- Study ESP systems, power transmission, single line diagram (SLD) interpretation and circuit breaker specifications.
- Apply electrical safety practice including arc flash awareness, bonding and grounding.
- Completed Chevron compliance training in Operational Excellence, Cyber Security, Data Privacy, Business Conduct and Ethics, and Conflict of Interest via Workday.

Research and Development Intern (SIWES) · ACE Quanser Robotics Lab, Obafemi Awolowo University 2024
Ile-Ife, Nigeria

- Operated the QDrone quadrotor UAV, QCar autonomous ground vehicle and QArm 6-DOF manipulator using MATLAB and Simulink.
- Applied PID control, sensor feedback and real-time system analysis to live hardware rather than simulation alone.
- Delivered a technical orientation presentation on the Quanser Aero 2 covering system architecture, control principles and laboratory safety.

Embedded Systems and EEE Tutor · Astar Tutorials 2023–Present
Ile-Ife, Nigeria

- Teach circuit analysis, digital logic and microcontroller programming to engineering students.
- Design structured tutorial sessions, sequencing theory and hands-on exercises so students build working circuits.

PROJECTS

aniwe · Python, Streamlit, OpenRouter, Vision-language models Aug 2026

- Built an AI costume-rating booth in Python and Streamlit with ten original hosts, per-session mood and quirk variation, and vision-language scoring across seven criteria collapsed into a single ranked score.
- Designed three-tier text-to-speech degradation (ElevenLabs, offline Piper, browser speech) and local faster-whisper speech-to-text so a blocked network reduces quality without stopping the system.

github.com/ayomide-abilewa/aniwe

AI-Powered Smart Attendance System · Python, OpenCV, YuNet, SFace Feb – Mar 2026

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Built an AI-powered attendance system using YuNet face detection and SFace recognition with cosine similarity matching, delivering the computer-vision pipeline and a Flask REST API backend.

- Developed the React, TypeScript and Tailwind frontend and a PostgreSQL schema covering students, courses, enrolment, sessions and attendance records.
- Defended the system at a final-year group project review and was invited to develop an extended production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Live Video Pipe Anomaly Detection · Python, YOLOv8, YOLOv11, Weighted Boxes Fusion 2026 – present

- Developing a two-stage cascaded detection architecture for real-time pipe anomaly identification, targeting industrial inspection.
- Built a parallel ensemble of RGB and edge-enhanced YOLO models merged with Weighted Boxes Fusion to improve precision across varied lighting and surface conditions.

Multispectral Acquisition Device for Hydroponics · ESP32, GY-2561, GY-31, I2C 2025

- Built a multi-sensor ESP32 acquisition node measuring light intensity and colour spectrum with the GY-2561 and GY-31, streaming live to a Blynk IoT Cloud dashboard.
- Integrated multiple I2C sensors on a single ESP32, handling bus arbitration and sensor calibration.

Smart Temperature-Controlled Fan · Arduino, C++, PWM, NTC thermistor 2024

- Led a team building a closed-loop fan speed controller on Arduino, driving PWM output from real-time NTC thermistor feedback.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded & Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI & Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals), Weighted Boxes Fusion

Instrumentation & Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20 mA and 3–15 PSI standards

Electrical Systems: SLD interpretation, Circuit breaker sizing, ESP systems, Power transmission, Arc flash safety, Hazardous area classification

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

LEADERSHIP AND TEACHING

SPAW Project Lead and Syllabus Team Lead · IEEE OAU Student Branch 2024–Present

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.
- Design all lesson plans, tutor guides and laboratory exercises for the programme.
- Produce structured PDF session materials including circuit diagrams, code examples and troubleshooting guides.

Core Instructor · Circuit Zero to Hero (ZTH) 2023–Present

- Run practical circuit design and electronics workshops for students with no prior background in the subject.

Editorial Team Member · EEE Students Society (EEESS), OAU 2023–Present

- Contribute to editorial output for the Electronic and Electrical Engineering Students Society.

Content Strategist · EEE Department Social Media Committee (TikTok / Instagram), OAU 2024–Present

Plans content for the department's social channels.

Campus Ambassador · Cowrywise 2023–Present

- Advocate financial literacy on campus as a Cowrywise campus ambassador.

TRAINING AND CERTIFICATIONS

- Operational Excellence Training. Chevron, 2025.
- Cyber Security and Data Privacy Awareness. Chevron, 2025.
- Business Conduct, Ethics and Conflict of Interest. Chevron, 2025.

- FPGA Design and HDL/Verilog Workshop (SWEP). Faculty of Engineering, Obafemi Awolowo University, 2024.
- Robotics Lab Orientation and Operations. ACE Quanser Lab, Obafemi Awolowo University, 2024.